Exploratory Data Analysis
Titanic Dataset

OBJECTIVE
Explore passenger data using statistics and visualizations.

TOOLS
Python, Pandas, Matplotlib, Seaborn, Google Colab

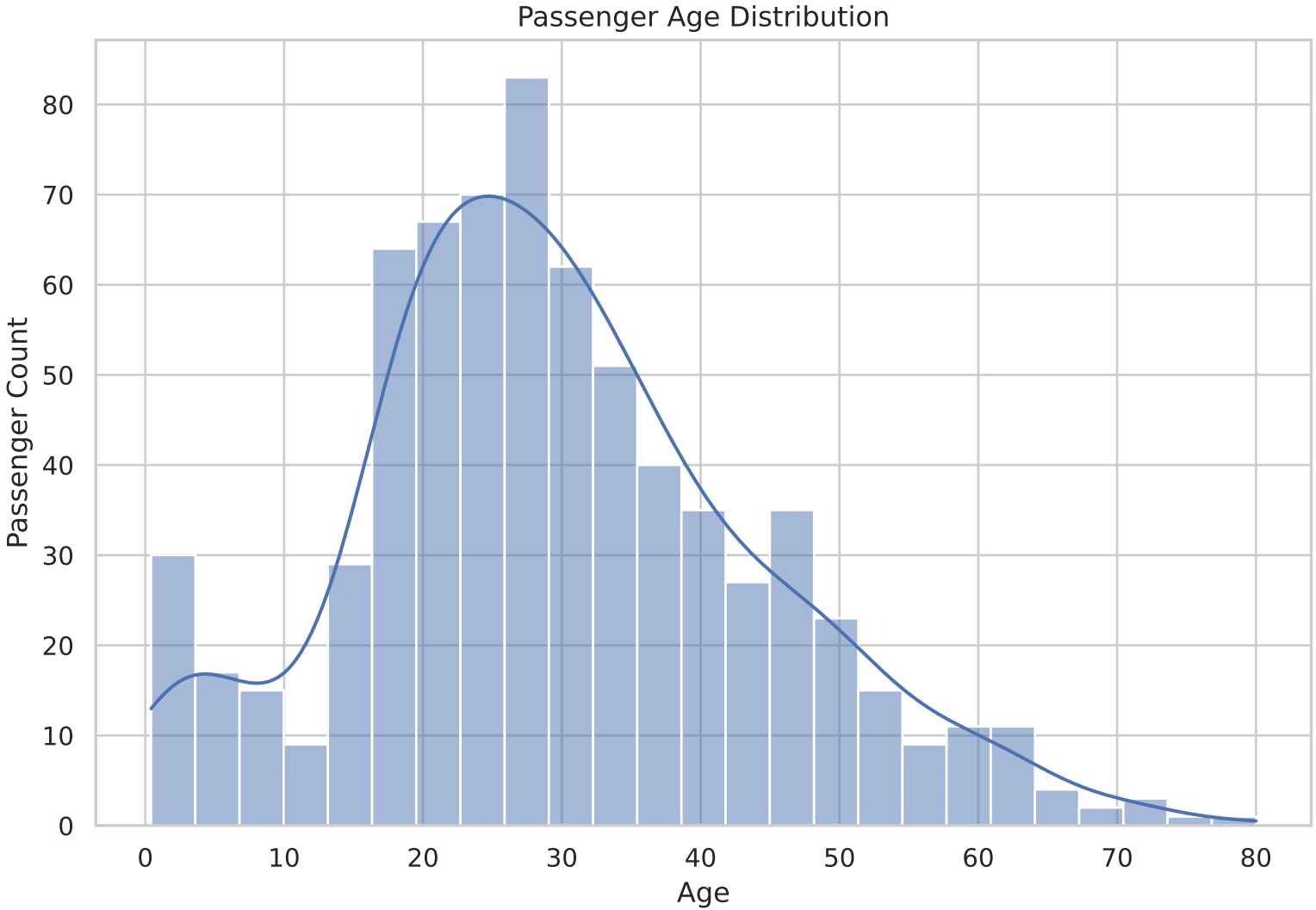
DATASET OVERVIEW
Rows: 891
Columns: 15

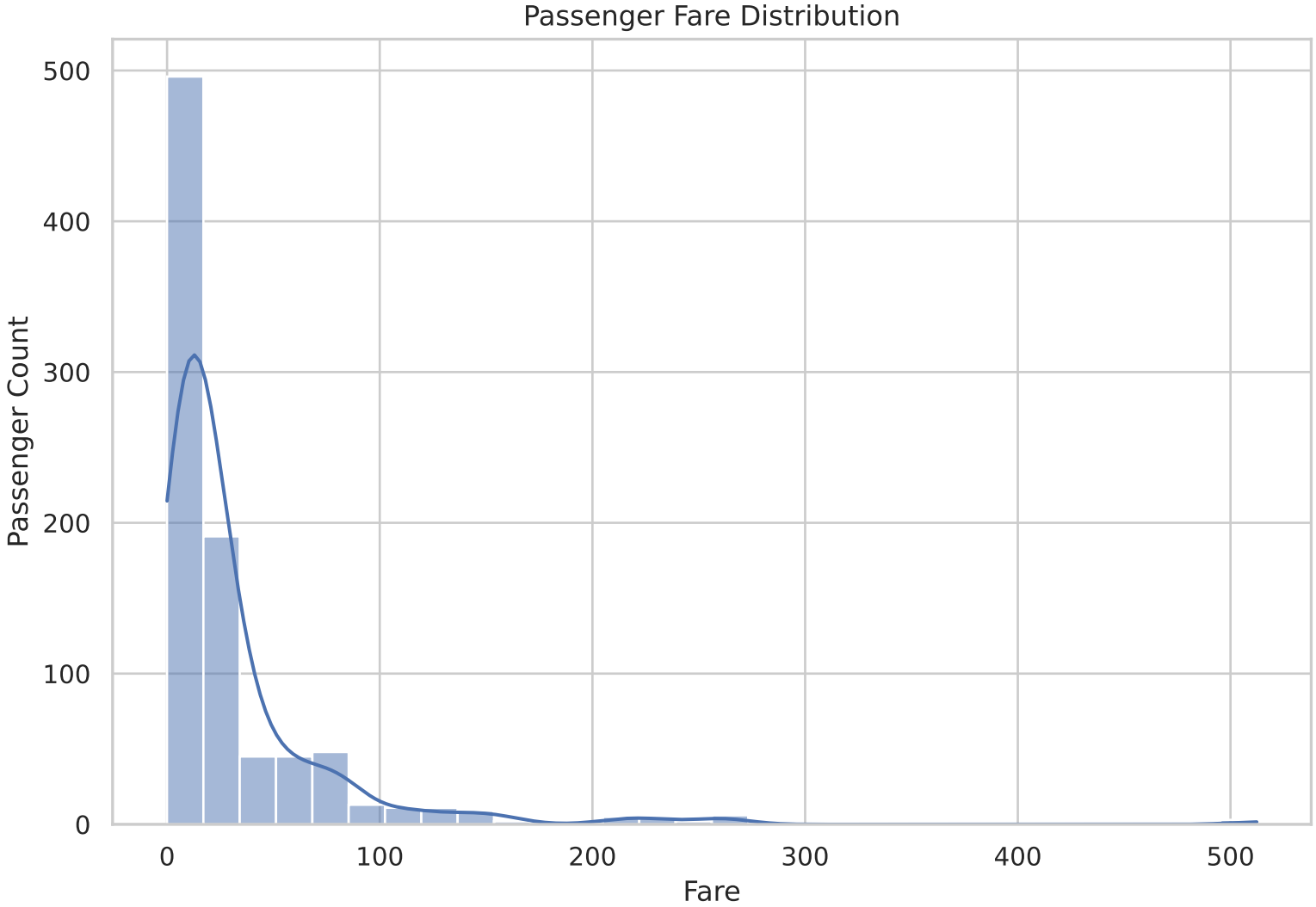
OVERALL SURVIVAL RATE
38.38%

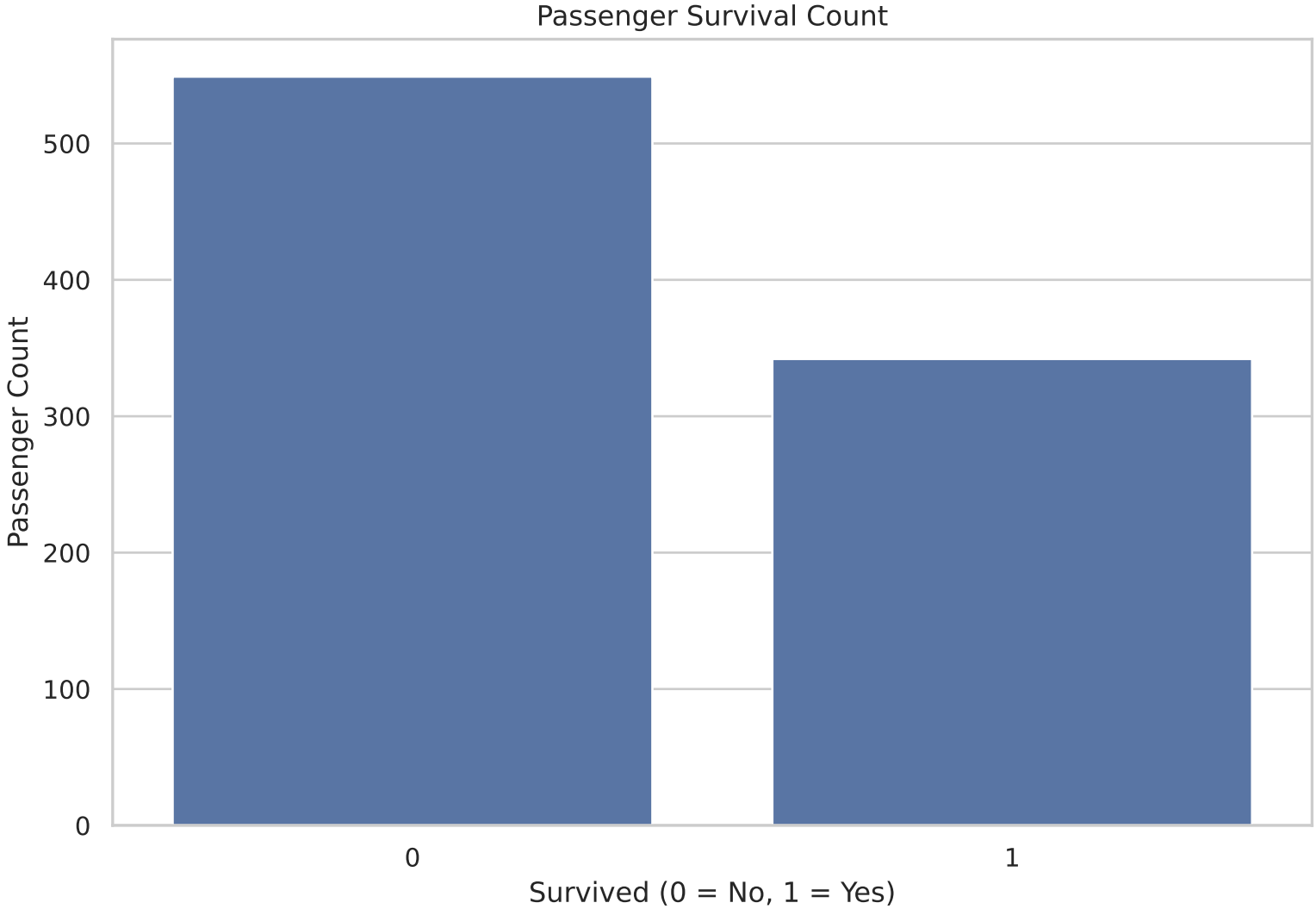
MISSING VALUES
869 missing cells across the dataset.

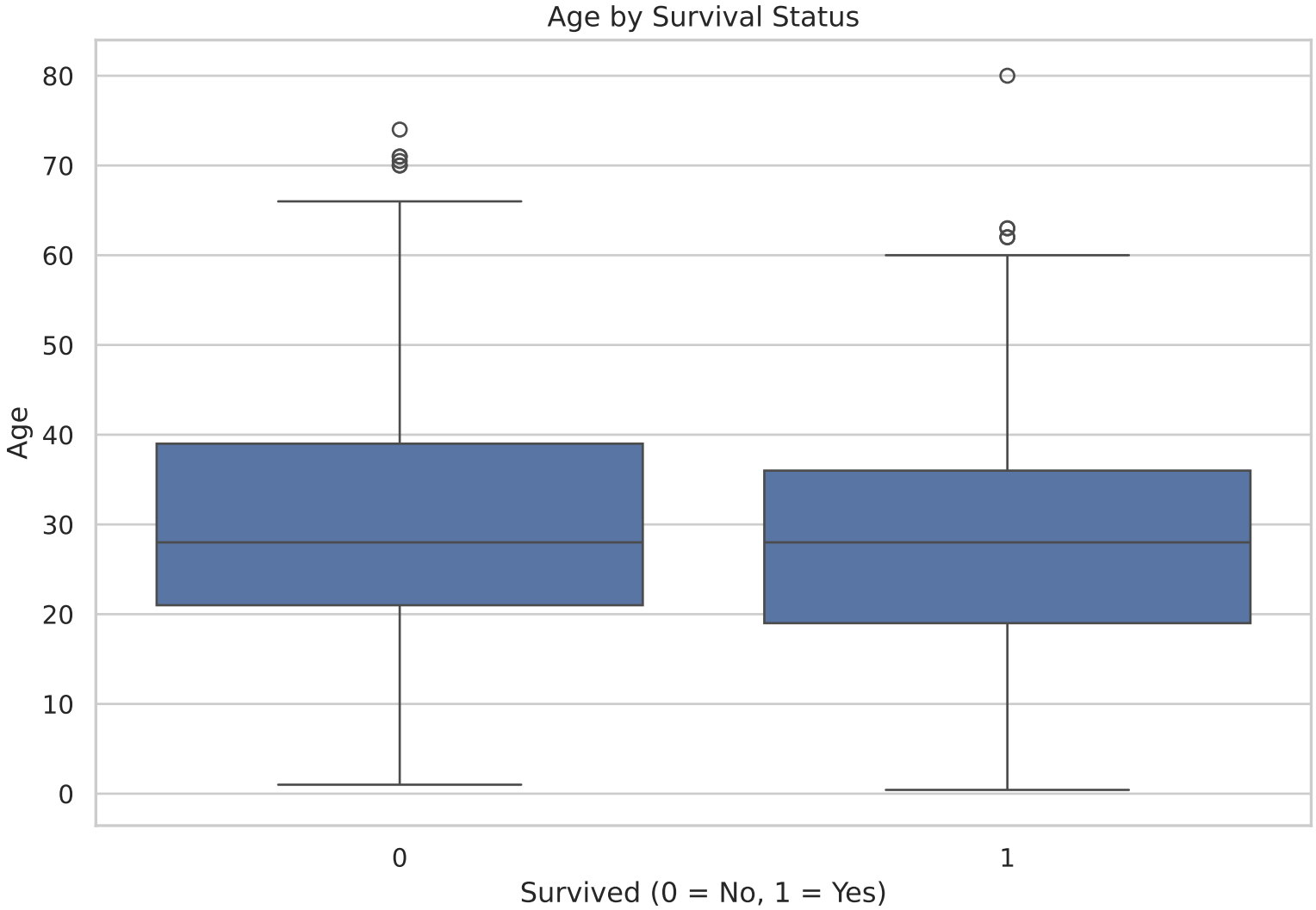
KEY ANALYSIS
- Passenger age and fare distributions
- Survival counts and survival rates
- Differences between passenger classes
- Relationships among numeric variables

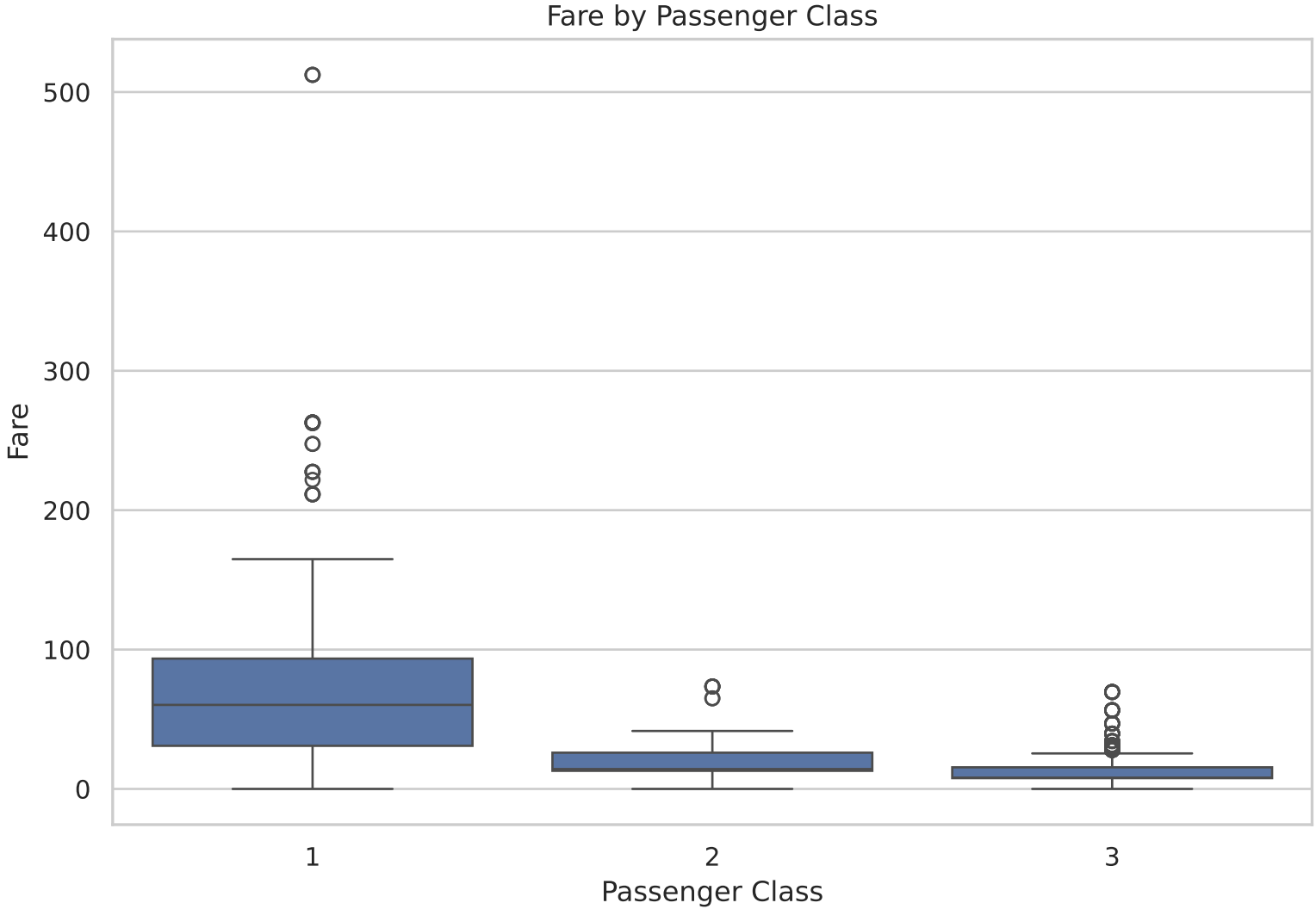
CONCLUSION
EDA helps identify patterns, distributions, missing data,
and associations that can guide further investigation.

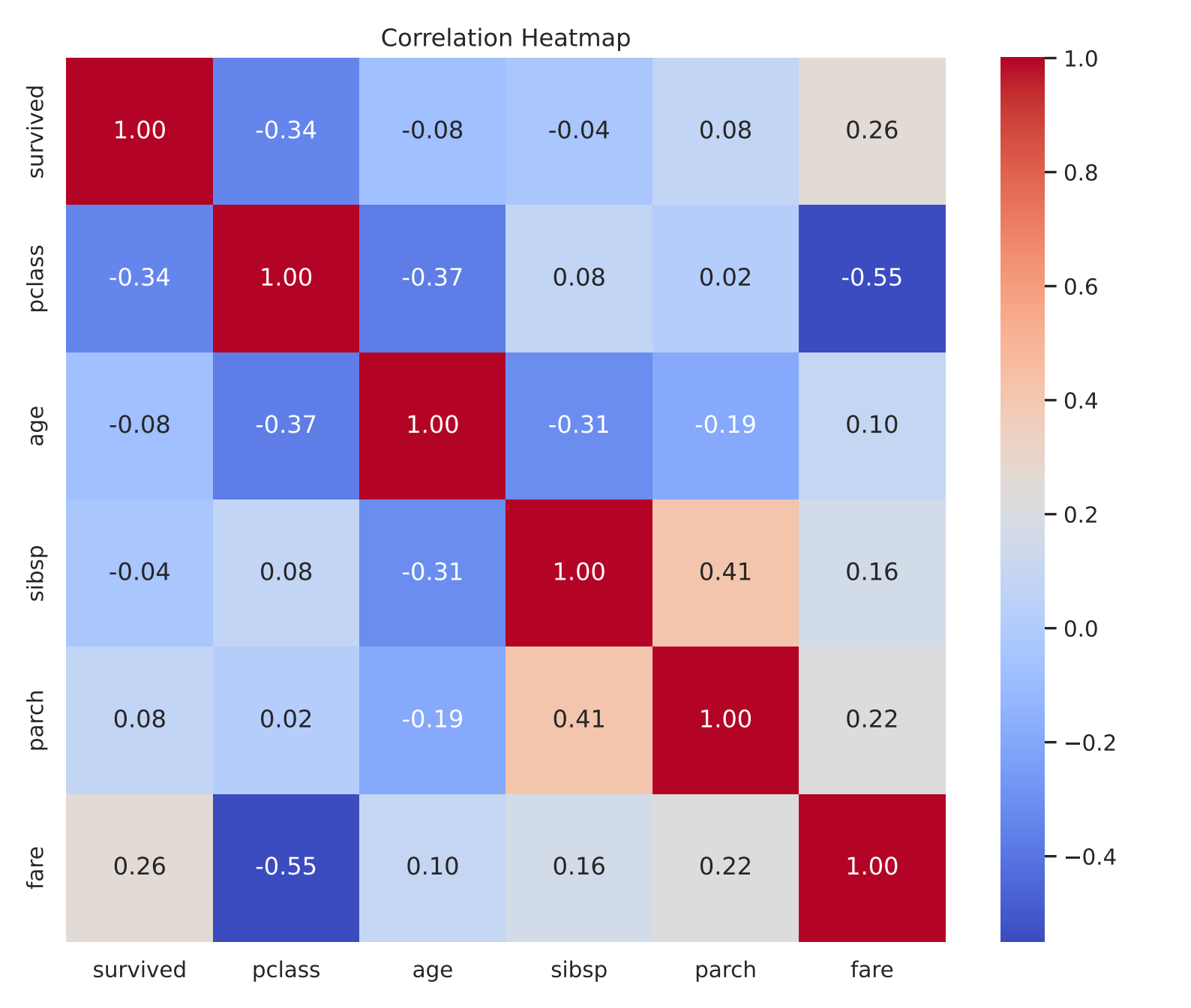

Passenger Age Distribution
Passenger Count
0
10
20
30
40
50
60
70
80
Age


Passenger Fare Distribution
Passenger Count
0
100
200
300
400
500
Fare
0
100
200
300
400
500


Passenger Survival Count
Passenger Count
0
100
200
300
400
500
0
1
Survived (0 = No, 1 = Yes)


Age by Survival Status
Age
80
70
60
50
40
30
20
10
0
0
1
Survived (0 = No, 1 = Yes)


Fare by Passenger Class
Fare
500
400
300
200
100
0
1
2
3
Passenger Class


Correlation Heatmap
survived
pclass
age
sibsp
parch
fare
1.00 -0.34 -0.08 -0.04 0.08 0.26
-0.34 1.00 -0.37 0.08 0.02 -0.55
-0.08 -0.37 1.00 -0.31 -0.19 0.10
-0.04 0.08 -0.31 1.00 0.41 0.16
0.08 0.02 -0.19 0.41 1.00 0.22
0.26 -0.55 0.10 0.16 0.22 1.00
survived
pclass
age
sibsp
parch
fare
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4